Four-Winding Transformer

Circuit Model and Governing Equations

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1 Introduction

This document derives the governing equations of a four-winding transformer sharing a common magnetic core. The four windings are grouped as two *primary* (indices 1, 2) and two *secondary* (indices 3, 4) windings. All windings are magnetically coupled through the same flux path in the core.

Notation used throughout:

- N_k — number of turns of winding k
- v_k, i_k, λ_k — terminal voltage, current, and flux linkage of winding k

- R_k — winding resistance
- L_{kk} — self-inductance of winding k
- $M_{jk} = M_{kj}$ — mutual inductance between windings j and k
- $\mathcal{R}$ — core reluctance
- Φ_m — common magnetising flux

2 Flux Linkage Matrix

Each winding concatenates the total core flux. The flux linkage of winding k is:

$$\lambda_k = L_{kk}\, i_k + \sum_{j \neq k} M_{kj}\, i_j, \qquad k = 1, 2, 3, 4. \tag{1}$$

In compact matrix form:

$$\boldsymbol{\lambda} = \mathbf{L}\,\mathbf{i}, \tag{2}$$

where

$$\begin{bmatrix} \lambda_1 \\ \lambda_2 \\ \lambda_3 \\ \lambda_4 \end{bmatrix} = \begin{bmatrix} L_{11} & M_{12} & M_{13} & M_{14} \\ M_{12} & L_{22} & M_{23} & M_{24} \\ M_{13} & M_{23} & L_{33} & M_{34} \\ M_{14} & M_{24} & M_{34} & L_{44} \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \\ i_3 \\ i_4 \end{bmatrix}. \tag{3}$$

The inductance matrix $\mathbf{L}$ is *symmetric* and *positive definite*.

2.1 Ideal Core — Perfect Coupling

For a lossless, infinite-permeability core (no leakage), all mutual inductances satisfy:

$$M_{jk} = \frac{N_j N_k}{\mathcal{R}} = \sqrt{L_{jj}\, L_{kk}}, \tag{4}$$

and the coupling coefficient between any pair equals unity:

$$k_{jm} \triangleq \frac{M_{jm}}{\sqrt{L_{jj}\, L_{mm}}} = 1. \tag{5}$$

3 Voltage Equations (KVL)

Applying Faraday's law and Kirchhoff's voltage law to each winding:

$$\boxed{v_k = R_k\, i_k + \frac{d\lambda_k}{dt} = R_k\, i_k + L_{kk}\frac{di_k}{dt} + \sum_{j \neq k} M_{kj}\frac{di_j}{dt}}, \qquad k = 1, 2, 3, 4. \tag{6}$$

In vector–matrix notation:

$$\mathbf{v} = \mathbf{R}\,\mathbf{i} + \mathbf{L}\,\frac{d\mathbf{i}}{dt}, \tag{7}$$

where $\mathbf{R} = \mathrm{diag}(R_1, R_2, R_3, R_4)$.

4 T-Equivalent Model with Leakage and Magnetising Inductances

4.1 Inductance Decomposition

Each self-inductance is decomposed into a *leakage* term $L_{\sigma k}$ and a *magnetising* term referred to winding k:

$$L_{kk} = L_{\sigma k} + \frac{N_k^2}{\mathcal{R}} = L_{\sigma k} + L_{m,k}, \tag{8}$$

where $L_{m,k} = N_k^2/\mathcal{R}$ is the magnetising inductance seen from winding k.

4.2 Magnetising Flux and Back-EMF

The common magnetising flux is driven by the total magnetomotive force (MMF):

$$\Phi_m = \frac{\mathcal{F}_{tot}}{\mathcal{R}} = \frac{1}{\mathcal{R}} \sum_{k=1}^{4} N_k \, i_k. \tag{9}$$

The back-EMF induced in winding k by the common flux is:

$$e_k = N_k \, \frac{d\Phi_m}{dt}. \tag{10}$$

4.3 Voltage Equations — T Form

Each winding equation splits into a resistive-leakage drop and a back-EMF:

$$v_k = R_k \, i_k + L_{\sigma k} \frac{di_k}{dt} + e_k. \tag{11}$$

4.4 Back-EMF Proportionality

Since $e_k = N_k \, d\Phi_m/dt$, the back-EMFs are proportional to the turns ratio with respect to any reference winding (here winding 1):

$$\frac{e_1}{N_1} = \frac{e_2}{N_2} = \frac{e_3}{N_3} = \frac{e_4}{N_4} \quad \Longrightarrow \quad e_k = \frac{N_k}{N_1} \, e_1. \tag{12}$$

5 Ampère's Law — MMF Balance

5.1 General Form

The net MMF sustains the core flux against the reluctance:

$$\sum_{k=1}^{4} N_k \, i_k = \mathcal{R} \, \Phi_m. \tag{13}$$

5.2 Ideal Core ($\mathcal{R} \to 0$, $i_m \approx 0$)

When the magnetising current is negligible compared to the load currents, the constraint reduces to:

$$\boxed{N_1 \, i_1 + N_2 \, i_2 + N_3 \, i_3 + N_4 \, i_4 = 0.} \tag{14}$$

This is the generalised Ampère balance for the four-winding ideal transformer.

6 Turns Ratios and Port Quantities

6.1 Definition

With winding 1 chosen as the voltage reference:

$$a_k \triangleq \frac{N_k}{N_1}, \qquad k = 2, 3, 4. \tag{15}$$

6.2 No-Load Voltages

Open-circuit terminal voltages obey:

$$V_k^{(\mathrm{oc})} = a_k \, V_1, \qquad k = 2, 3, 4. \tag{16}$$

6.3 Current Constraint (Ideal Core)

Substituting the turns ratios into (14):

$$i_1 + a_2 \, i_2 = -(a_3 \, i_3 + a_4 \, i_4). \tag{17}$$

The left-hand side represents the combined primary excitation; the right-hand side is the reflected secondary load.

7 Summary of Parameters

Table 1: Parameter summary for the four-winding transformer model.

Symbol	Unit	Description
$L_{\sigma k}$	H	Leakage inductance of winding k
$L_{m,k} = N_k^2/\mathcal{R}$	H	Magnetising inductance seen from winding k
$M_{jk} = a_j a_k L_{m,1}$	H	Mutual inductance between windings j and k
R_k	Ω	Winding resistance
$\mathcal{R}$	A/Wb	Core reluctance (includes air gap if present)
$a_k = N_k/N_1$	—	Turns ratio with respect to winding 1
e_k	V	Back-EMF induced by common core flux
Φ_m	Wb	Common magnetising flux

8 Remarks on Multi-Port Applications

- The inductance matrix $\mathbf{L}$ in (3) is the natural starting point for multi-active-bridge (MAB) converter analysis, where the leakage inductances $L_{\sigma k}$ act as power-transfer reactances.
- For a dual-active-bridge (DAB) extended to four ports, ZVS conditions for each switch must be verified independently for each winding current at the switching instants.
- When two primary windings are fed from different voltage sources (e.g. two AFE converters), circulating currents between them are governed by $(v_1 - v_2)/(L_{\sigma 1} + L_{\sigma 2} + M_{12,\mathrm{leak}})$ and must be controlled or suppressed by design.

- The constraint (14) allows reduction of the state space: given three independent currents, the fourth is determined. This is exploited in state-space models for simulation and control design.